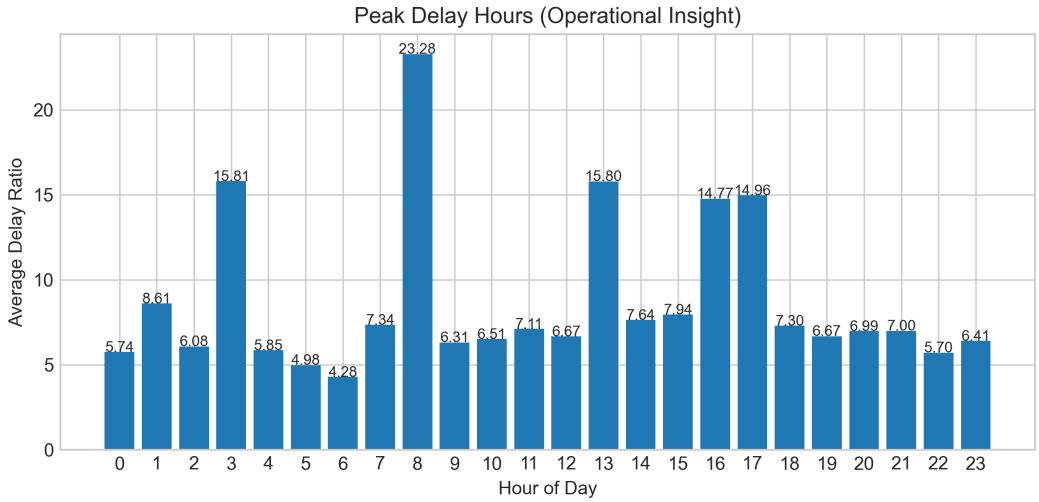
AI-Powered Trip Efficiency Report

Key Metrics

Average Delay Ratio represents the overall efficiency of trips by measuring how much delay exists compared to expected time.
Value: 8.87
High Delay Trips counts the number of trips that exceeded the acceptable delay threshold, highlighting operational issues and outliers.
Value: 134841
Average Speed indicates the overall movement efficiency of trips and reflects traffic conditions and routing performance.
Value: 14.41 km/h
Model MAE (Mean Absolute Error) measures the average prediction error of the model in seconds, indicating forecasting accuracy.
Value: 413.88 seconds

Peak Delay Hours



This chart shows average delay ratio per hour of the day. Higher values indicate peak congestion periods where trip efficiency drops significantly.

Top 7 Problematic Trips

ID	Vendor	Delay Ratio	Prediction Deviation
id0053347	1	1476.62	3524066.41
id0369307	1	2854.97	1228911.25
id1325766	1	11352.89	556771.86
id1864733	1	802.26	290673.14
id0759846	2	1119547.83	86063.19
id0067152	2	1073.65	86017.8
id3762091	2	1330.06	86010.64

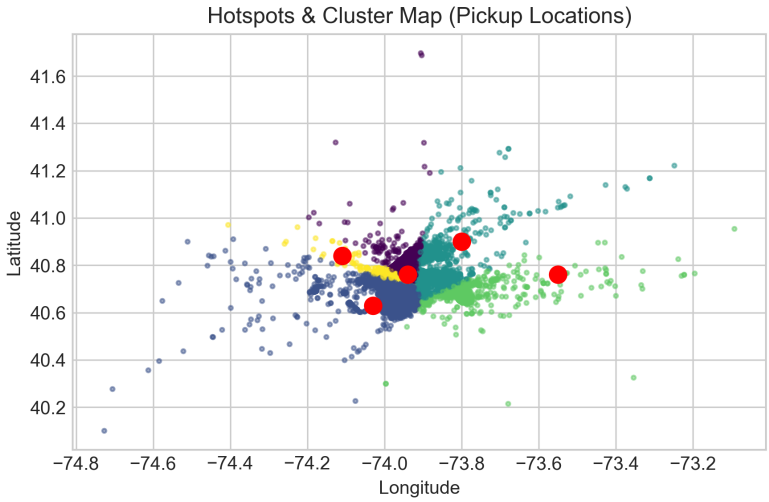
High Delay Zones (Critical Areas)

Zone 40.63_-74.03: Delay Ratio 17599.55
Zone 40.76_-73.55: Delay Ratio 2730.5
Zone 40.9_-73.8: Delay Ratio 2429.55
Zone 40.76_-73.94: Delay Ratio 2367.87
Zone 40.84_-74.11: Delay Ratio 2357.52

Geographical Clusters Performance

Cluster 3: Delay Ratio 48.1
Cluster 0: Delay Ratio 12.12
Cluster 2: Delay Ratio 9.68
Cluster 1: Delay Ratio 7.29
Cluster 4: Delay Ratio 6.03

Geographical Hotspots & Clusters



This map highlights high-demand and high-delay zones using clustering and grid-based segmentation. Red areas indicate critical zones requiring route or supply optimization.

Captain Performance Ranking

Vendor 1.0 → Score: 0.3
Vendor 2.0 → Score: 0.24
This ranking system evaluates vendors based on delay efficiency, prediction accuracy, and speed performance. Higher scores represent more reliable and efficient captains.

AI Recommendations

- Increase incentives during peak delay hours
- Optimize routing for congested zones
- Rebalance captain supply